

Mathematical Induction

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I Introduction

When writing a proof that has to do with the set of natural numbers, $\mathbb{N}$, we can use *induction* as our preferred method of proof. First, we formally define The Principle of Induction.

The Principle of Induction

Let $n \in \mathbb{N}$ and $S \subset \mathbb{N}$ that has the following properties:

- i. $1 \in S$
- ii. $\forall n \in S$, then, $n + 1 \in S$

Which gives that $S = \mathbb{N}$.

When starting a proof by induction, we first check the base case. Often, this is $n = 1$ (but it could be any natural number, say if the identity is true for $n \geq 5$, then our base case can start at $n = 5$). Once we have shown that the identity is true for at least 1 natural number, we assume that it is true for any natural number, n . To prove this, we want to show that it is also true for $n + 1$. The idea that it is true for $n + 1$ is called the inductive hypothesis (or inductive step). Once we have shown that our inductive hypothesis is true, we are finished and the proof is complete.

[†]Definitions and example problems from MATH 42001: Analysis I instructed by Dr. Artem Zvavitch, *Introduction to Real Analysis* by Robert G. Bartle and Donald R. Sherbert, and *How to Prove It* by Daniel J. Velleman

II Examples

We will start with a very simple example.

1. Prove that $2^0 + 2^1 + \dots + 2^n = 2^{n+1} - 1, \forall n \in \mathbb{N}$

Proof. We start by checking the base case $n = 1$:

$$2^0 + 2^1 = 3 = 2^2 - 1$$

Inductive step: Assume that $2^0 + 2^1 + \dots + 2^n = 2^{n+1} - 1, \forall n \in \mathbb{N}$. (We want to show that $2^0 + 2^1 + \dots + 2^n + 2^{n+1} = 2^{n+2} - 1$) Then,

$$(2^0 + 2^1 + \dots + 2^n) + 2^{n+1} = 2^{n+1} - 1 + 2^{n+1} \implies$$

$$2^0 + 2^1 + \dots + 2^n + 2^{n+1} = 2 \cdot 2^{n+1} - 1 \implies$$

$$2^0 + 2^1 + \dots + 2^n + 2^{n+1} = 2^{n+2} - 1$$

Therefore, by induction, $2^0 + 2^1 + \dots + 2^n = 2^{n+1} - 1, \forall n \in \mathbb{N}$. $\square$

One strategy we can always try with induction is adding a $n+1$ term to both sides, typically one that matches the structure of the left hand side. In this case, it was 2^{n+1} .

2. Prove that $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \geq \sqrt{n}, \forall n \in \mathbb{N}$

Proof. Base case $n = 1$:

$$\frac{1}{\sqrt{1}} = 1 = \sqrt{1}$$

Assume that $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \geq \sqrt{n}, \forall n \in \mathbb{N}$. Then,

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n} + \frac{1}{\sqrt{n+1}} \implies$$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n} + \frac{1}{\sqrt{n+1}} - \left(\sqrt{n+1} - \sqrt{n} \right) \implies$$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n} + \frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n+1} + \sqrt{n}} \implies$$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$$

Therefore, $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \geq \sqrt{n}, \forall n \in \mathbb{N}$. $\square$

Now we will do a more complicated proof.

3. Prove that $2^n \geq n^2, \forall n \geq 5$, where $n \in \mathbb{N}$

Proof. Base case $n = 5$:

$$2^5 = 32 \geq 5^2 = 25$$

Assume that $2^n \geq n^2, \forall n \geq 5$, where $n \in \mathbb{N}$. In this case adding a $n + 1$ term doesn't do much so we will try and multiply both sides by 2 since that will get us 2^{n+1} on the left hand side.

$$\begin{aligned} 2 \cdot 2^n &\geq n^2 \cdot 2 \implies \\ 2^{n+1} &\geq 2n^2 \implies \\ 2^{n+1} &\geq n^2 + n^2 \end{aligned}$$

From here it is not exactly clear what to do next. So, we will use a trick and note that since $n \geq 5 \implies n^2 \geq 5n$. Our goal is to try and manipulate $n^2 + n^2$, so we can observe that $n^2 + n^2 \geq n^2 + 5n$. We can now put this into our inequality above in order to manipulate the right hand side to get $(n + 1)^2$:

$$\begin{aligned} 2^{n+1} &\geq n^2 + n^2 \geq n^2 + 5n \implies \\ 2^{n+1} &\geq n^2 + n^2 \geq n^2 + 2n + 3n \text{ (so we can factor)} \implies \end{aligned}$$

Now, we can note that $(n + 1)^2 = n^2 + 2n + 1$. So we can see that:

$$2^{n+1} \geq n^2 + n^2 \geq n^2 + 2n + 3n > n^2 + 2n + 1 = (n + 1)^2$$

So, by our inequality,

$$2^{n+1} \geq (n + 1)^2$$

Therefore, $2^n \geq n^2, \forall n \geq 5$, where $n \in \mathbb{N}$. $\square$

Finally, we will do a proof involving summations.

4. Prove that $\sum_{k=1}^n \frac{a-1}{a^k} = 1 - \frac{1}{a^n}, \forall a \neq 0$

Proof. Base case $k, n = 1$:

$$\frac{a-1}{a^1} = 1 - \frac{1}{a^1} = 1 - \frac{1}{a^1}$$

Assume $\sum_{k=1}^n \frac{a-1}{a^k} = 1 - \frac{1}{a^n}, \forall a \neq 0$. Then,

$$\begin{aligned} \sum_{k=1}^n \frac{a-1}{a^k} + \frac{a-1}{a^{n+1}} &\implies \\ 1 - \frac{1}{a^n} + \frac{a-1}{a^{n+1}} - \frac{1}{a^{n+1}} &\left(\frac{a-1}{a^k} = 1 - \frac{1}{a^n} \text{ from inductive step} \right) \implies \\ &1 - \frac{1}{a^{n+1}} \end{aligned}$$

Therefore, $\sum_{k=1}^n \frac{a-1}{a^k} = 1 - \frac{1}{a^n}, \forall a \neq 0$. $\square$

The trick here is to use what we assumed in the inductive step to rewrite one part of the summation. That way, everything is in the same terms and is much easier to work with.